

A one-big-jump counterexample to the $p < 2$ subcritical moderate-deviation extension

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Status. Candidate counterexample, full negative resolution of the final question in Remark 1.1(c) of arXiv:2003.11442, subject to human review.

1 Source question

Alonso-Gutiérrez, Prochno, and Thäle [1] prove a quadratic moderate-deviation principle (MDP) for normalized Euclidean norms of random orthogonal projections of uniform points in ℓ_p^n -balls when $p \geq 2$. At the end of Remark 1.1(c), page 5, they ask:

“Another open problem is to decide whether or not the rate function in the MDP in Theorem C is the same also for $1 \leq p < 2$.”

LDP, MDP AND THE KLS CONJECTURE

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- (a) Although the results we presented so far were formulated for uniform random vectors X_n in $\mathbb{B}_p^n$, we will prove more general versions for random vectors that are distributed in $\mathbb{B}_p^n$ according to a distribution from the class introduced in [5]. As special cases this includes the uniform probability measure or the cone probability measure on $\mathbb{B}_p^n$, but also image measures of these measures on higher-dimensional ℓ_p^n -balls under coordinate projections. We refer to the discussion in the next section.
- (b) We would like to point out that the constant $\alpha_{p,\lambda}$ in [2] coincides with $\frac{1}{2\sigma^2(p,\lambda)}$, where $\sigma^2(p,\lambda)$ is the variance of the centered Gaussian random variable appearing as the limit in the central limit theorem for $\mathcal{X}_{n,p}$ from [3]. In other words, this means that the rate function $\mathbb{I}(y)$ in Theorem C can be rewritten as $\mathbb{I}(y) = \frac{y^2}{2\sigma^2(p,\lambda)}$, as expected.
- (c) We leave it as an open problem for future research to study moderate deviations at the supercritical scales $t_n = \omega(\sqrt{k_n})$ (and $t_n = o(\sqrt{n})$). In addition, it would be interesting to investigate the situation where the involved parameters neither satisfy $t_n = o(\sqrt{n - k_n})$ nor $(n - k_n) = o(\frac{\sqrt{k_n}}{t_n})$. Another open problem is to decide whether or not the rate function in the MDP in Theorem C is the same also for $1 \leq p < 2$.

2. PRELIMINARIES

Figure 1: The open question in Remark 1.1(c) of the source paper, page 5.

For a uniform random vector X_n in B_p^n , an independent Haar-uniform k_n -dimensional subspace E_n , and

$$m_p = M_p(2) = p^{2/p} \frac{\Gamma(3/p)}{\Gamma(1/p)},$$

the source normalization is

$$\mathcal{X}_{n,p} = \frac{n^{1/p}}{\sqrt{m_p}} \|P_{E_n} X_n\|_2 - \sqrt{k_n}.$$

Theorem C assumes $t_n \rightarrow \infty$, $t_n = o(\sqrt{k_n})$, and either $t_n = o(\sqrt{n - k_n})$ or the alternative endpoint condition, and concludes that $\mathcal{X}_{n,p}/t_n$ has speed t_n^2 and a strictly positive quadratic rate function.

2 Idea of the proof

The restriction $p \geq 2$ is structural. In the exact representation used by the source, the squared p -generalized Gaussian coordinate Z_i^2 occurs in the numerator. When $p < 2$, it has stretched-exponential tail $\exp(-cu^{p/2})$. One coordinate of size $Z_1^2 \asymp t_n \sqrt{n}$ already moves $\mathcal{X}_{n,p}$ by order t_n . The logarithmic cost of this event is only $(t_n \sqrt{n})^{p/2}$. At the explicit admissible scale $t_n = n^{1/(4-p)}$, this is $o(t_n^2)$. Consequently a fixed positive upper tail has rate zero on the proposed speed, contradicting every positive quadratic MDP rate.

3 Counterexample theorem

Theorem 1. Fix $1 \leq p < 2$. Let $k_n = \lfloor n/2 \rfloor$ and

$$t_n = n^{1/(4-p)}.$$

Let X_n be uniform on B_p^n , let E_n be an independent Haar-uniform k_n -dimensional subspace, and define $\mathcal{X}_{n,p}$ as above. Then

$$\lim_{n \rightarrow \infty} \frac{1}{t_n^2} \log \mathbb{P} \left(\frac{\mathcal{X}_{n,p}}{t_n} \geq 1 \right) = 0.$$

In particular, $\mathcal{X}_{n,p}/t_n$ does not satisfy an MDP with speed t_n^2 and the quadratic rate function proposed in Theorem C. Hence the final question in Remark 1.1(c) has a negative answer for every $1 \leq p < 2$.

Proof. Put $d_n = t_n/\sqrt{n}$. Since $p < 2$,

$$t_n \rightarrow \infty, \quad d_n \rightarrow 0,$$

and, because $k_n/n \rightarrow 1/2$, both $t_n = o(\sqrt{k_n})$ and $t_n = o(\sqrt{n - k_n})$ hold. Thus these scales satisfy the hypotheses in the question.

Step 1: the exact representation. Let $g_1, \dots, g_n$ be independent standard Gaussians, let $Z_1, \dots, Z_n$ be independent p -generalized Gaussians with density

$$f_p(z) = \frac{\exp(-|z|^p/p)}{2p^{1/p}\Gamma(1+1/p)},$$

and let W be exponential with rate $1/p$, all independent. Equation (3.1) and item (i) on page 5 of [1] give

$$\frac{n^{1/p}}{\sqrt{k_n}} \|P_{E_n} X_n\|_2 \stackrel{d}{=} G_n \frac{(n^{-1} \sum_{i=1}^n Z_i^2)^{1/2}}{(n^{-1} \sum_{i=1}^n |Z_i|^p + W/n)^{1/p}},$$

where

$$G_n = \frac{(k_n^{-1} \sum_{i=1}^{k_n} g_i^2)^{1/2}}{(n^{-1} \sum_{i=1}^n g_i^2)^{1/2}}.$$

Since $\mathbb{E}Z_1^2 = m_p$ and $\mathbb{E}|Z_1|^p = 1$, define

$$R_n = \left(\frac{n^{-1} \sum Z_i^2}{m_p} \right)^{1/2} \left(n^{-1} \sum |Z_i|^p + \frac{W}{n} \right)^{-1/p}.$$

Then

$$\mathcal{X}_{n,p} \stackrel{d}{=} \sqrt{k_n} (G_n R_n - 1). \quad (1)$$

Step 2: a deterministic one-big-jump event. Set $L = 12m_p$, and choose a sufficiently small fixed $\eta > 0$. Consider the intersection A_n of the following events:

$$Lt_n \sqrt{n} \leq Z_1^2 \leq (L+1)t_n \sqrt{n}, \quad (A1)$$

$$\left| \sum_{i=2}^n (Z_i^2 - m_p) \right| \leq \eta t_n \sqrt{n}, \quad (A2)$$

$$\left| \sum_{i=2}^n (|Z_i|^p - 1) \right| \leq \eta t_n \sqrt{n}, \quad (A3)$$

$$W \leq \eta t_n \sqrt{n}, \quad (A4)$$

$$G_n \geq 1 - \eta d_n. \quad (A5)$$

On this event, for all sufficiently large n ,

$$\frac{1}{n} \sum_{i=1}^n Z_i^2 \geq m_p + (L - 2\eta) d_n. \quad (2)$$

Moreover,

$$\frac{Z_1^p}{n d_n} \leq (L+1)^{p/2} (t_n \sqrt{n})^{p/2-1} \longrightarrow 0,$$

because $p/2 - 1 < 0$. Hence (A3)–(A4) imply

$$\frac{1}{n} \sum_{i=1}^n |Z_i|^p + \frac{W}{n} \leq 1 + 3\eta d_n \quad (3)$$

for large n . Using $\sqrt{1+u} \geq 1 + u/3$ for small $u \geq 0$ and $(1+v)^{-1/p} \geq 1 - v/p$ for $v \geq 0$, equations (2)–(3) yield

$$G_n R_n \geq (1 - \eta d_n) \left(1 + \frac{L - 2\eta}{3m_p} d_n \right) \left(1 - \frac{3\eta}{p} d_n \right).$$

Because $L/(3m_p) = 4$, we may first choose η small and then n large so that

$$G_n R_n \geq 1 + 3d_n. \quad (4)$$

Combining (1) and (4),

$$\frac{\mathcal{X}_{n,p}}{t_n} \geq 3\sqrt{\frac{k_n}{n}} > 1$$

for all sufficiently large n .

Step 3: logarithmic cost. The variables in (A5), (A4), (A1), and (A2)–(A3) are independent across the indicated Gaussian, W , first-coordinate, and remaining-coordinate blocks. Since all moments of Z_i are finite, Chebyshev's inequality gives

$$\mathbb{P}((A2) \cap (A3)) \longrightarrow 1.$$

Also $\mathbb{P}(A4) \rightarrow 1$. Finally, $G_n - 1 = O_{\mathbb{P}}(n^{-1/2})$, whereas $d_n = t_n n^{-1/2}$ with $t_n \rightarrow \infty$, so $\mathbb{P}(A5) \rightarrow 1$.

Write $a_n = (Lt_n\sqrt{n})^{1/2}$ and $b_n = ((L+1)t_n\sqrt{n})^{1/2}$. The displayed density gives the elementary lower bound

$$\begin{aligned} \mathbb{P}(A1) &= 2 \int_{a_n}^{b_n} f_p(z) dz \\ &\geq c_p(b_n - a_n) \exp(-b_n^p/p), \end{aligned}$$

where $c_p > 0$. Therefore

$$\log \mathbb{P}(A_n) \geq -\frac{(L+1)^{p/2}}{p} (t_n\sqrt{n})^{p/2} + O(\log n + \log t_n) + o(1). \quad (5)$$

The chosen scale makes

$$\frac{(t_n\sqrt{n})^{p/2}}{t_n^2} = n^{(p-2)/4} \longrightarrow 0.$$

Dividing (5) by t_n^2 shows

$$\liminf_{n \rightarrow \infty} \frac{1}{t_n^2} \log \mathbb{P}\left(\frac{\mathcal{X}_{n,p}}{t_n} \geq 1\right) \geq 0.$$

The reverse inequality is automatic because probabilities are at most one, so the asserted limit is zero.

Step 4: contradiction to the proposed MDP. If $\mathcal{X}_{n,p}/t_n$ obeyed an MDP with speed t_n^2 and rate $I(x) = \alpha_{p,1/2}x^2$ with $\alpha_{p,1/2} > 0$, the large-deviation upper bound for the closed set $[1, \infty)$ would give

$$\limsup_{n \rightarrow \infty} \frac{1}{t_n^2} \log \mathbb{P}\left(\frac{\mathcal{X}_{n,p}}{t_n} \geq 1\right) \leq -\alpha_{p,1/2} < 0,$$

contradicting the limit just proved. □

4 Verification and scope

The algebraic checks are:

$$\frac{1}{4-p} < \frac{1}{2}, \quad \frac{1}{4-p} > \frac{p}{2(4-p)}, \quad \frac{(t_n\sqrt{n})^{p/2}}{t_n^2} = n^{(p-2)/4}.$$

Thus the example lies inside the source's subcritical scale window but above the Gaussian-to-one-big-jump crossover for every $p < 2$. The included script checks these identities numerically over a grid of p values. No numerical claim is used in the proof.

This result resolves only the final $p < 2$ question in Remark 1.1(c). It does not address the preceding questions about supercritical t_n or the gap between the two conditions relating t_n and $n - k_n$.

5 Novelty check

On 22 July 2026 we searched the run indexes and arXiv/web exact phrases for the source title, “same also for $1 \leq p < 2$ ”, “subcritical scales”, “ p -generalized Gaussian moderate deviations”, and random projections of ℓ_p^n -balls. No paper explicitly answering Remark 1.1(c) was found. Tugel–Gantert–Prochno [2] give a general 2026 description of Gaussian and few-big-jump regimes for stretched-exponential sums, but their paper neither cites arXiv:2003.11442 nor states this projected-norm counterexample. The proof above is self-contained once the exact representation from the source is granted. Novelty confidence is therefore medium-high; a reviewer should still run a citation-database search.

References

- [1] D. Alonso-Gutiérrez, J. Prochno, and C. Thäle, *Large deviations, moderate deviations, and the KLS conjecture*, arXiv:2003.11442 (2020).
- [2] P. Tugel, N. Gantert, and J. Prochno, *Large deviations for sums of multivariate stretched-exponential random variables: the few-big-jumps principle*, arXiv:2602.01168 (2026).